

3.2 Infinite Series

Definition: A **series** (specifically infinite series) results when we want to add up all the terms of a sequence.

For example:

$\{2, 4, 6, 8, 10, \dots\}$ is a sequence.

$2 + 4 + 6 + 8 + 10 + \dots$ is a series.

Notation (Sigma Notation):

Note: $\sum_{i=1}^n a_i$ is known as the n th partial sum of the series $\sum_{i=1}^{\infty} a_i$.

Definition:

Example 1: Find the first, second, third, and fourth partial sum of the following series.

$$1 - 1 + 1 - 1 + 1 - 1 + \dots$$

Does the series converge or diverge?

Example 2: Find the first, second, third, and fourth partial sums of the series: $\sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i$

Example 3: Show that the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ is convergent, and find its sum.

Definition: A series is called a geometric series if

For example:

Theorem:

Proof:

Example 4: Consider the number $0.\overline{9}$ (point 9 repeating).

a) Write this value as a geometric series in sigma notation.

b) Find the exact value of the series from part (a).

Example 5 (you try):

Determine whether the following series is convergent or divergent. If it is convergent find its exact sum:

$$\sum_{n=1}^{\infty} (8^{2n-1} \cdot 3^{3-4n})$$

Example 6:

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k}$ which is called the *harmonic series*. The n^{th} partial sum of the series is: $s_n = \sum_{k=1}^n \frac{1}{k}$.

- (a) Find s_5 (exactly).
- (b) Find s_{100} (rounded to four decimal places).
- (c) Find s_{200} (rounded to four decimal places).
- (d) Based upon your answers to parts (a) through (c), do you believe the harmonic series to be convergent or divergent? If you believe it is convergent, what do you think it converges to?

Theorem: The harmonic series is
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Proof: